of much more data in parallel, simultaneously, and all permutations of that data, to discover the best patterns that describe it. This will lead to fundamentally more powerful forms of AI, much more quickly than we expect. The speed at which the world develops AI will go up considerably with the power of QC. It helps in the following:

- Detecting statistical anomalies
- Finding compressed models
- Recognising images and patterns
- Training neural networks
- Verifying and validating software
- Classifying unstructured data
- Diagnosing circuit faults

Financial services: JPMorgan is working with IBM to explore how quantum computers can assist with trading strategies, portfolio optimisation, asset pricing and risk analysis. Similarly, Barclays is also participating in the IBM Q Network to investigate if quantum computers could be used to optimise the settlement of large batches of financial transactions. In addition, QC also helps in detecting market instabilities, optimising trading trajectories and asset pricing, as well as in hedging.

Healthcare and medicine: QC allows users to model complex molecular interactions at an atomic level. This helps in medical research, leading to accelerated drug discovery. Functions like drug design, molecular dynamics, fraud detection, optimising therapy treatments, and creating protein models while optimising the effects, can be addressed by QC.

QC helps in the analysis of drug interactions, finds cures for previously incurable diseases and helps to accelerate the time to market for new drugs.

QC simulations can help to design and choose the next generation of drugs and cancer cures. Research towards this is under way.

Air traffic management: In 2011, Lockheed Martin was the first purchaser of a quantum computer manufactured by D-Wave Systems and has continued to investigate the uses of the technology for application in air traffic management and system verification.

Airbus is similarly investigating how quantum computers could speed up its research activities and has invested in the quantum computing software company, QC Ware.

Transportation: Mobility as a Service (MaaS) helps to optimally distribute transportation capacities in a very dynamic urban system, where there are millions of transportation requests from people and for goods, at short time intervals. QC helps to make the transportation system highly reactive to disruptions.

QC helps in analysing the congestion caused by traffic accidents, commuter-train line breakdowns, extreme weather events, etc. This leads to the setting up of a dynamic transport system that handles the requests in real-time.

Challenges in quantum computing today

The following are the challenges in QC:

- Limited access to QC hardware and limited know-how on how to leverage the potential of QC hinder the development of quantum software.
- The lack of good software, i.e., more QC algorithms that solve real-world problems.
- Limited access to industrial manufacturing facilities makes it challenging to explore the scaling up of QC.
- Technological challenges like limited qubit connectivity, too low gate fidelities, or the large amounts of qubits required for error correction. Therefore, detecting, controlling and correcting of errors becomes a major challenge.
- Lack of collaboration and exchange between industry and academia.
- The inability to leverage superconductivity to create a qubit and maintain a quantum state. To work with these superconducting qubits for extended periods, requires them to be maintained at very low temperatures. Quantum computers operate at temperatures close to absolute zero, colder than the vacuum of space. Maintaining such a low temperature is a big challenge.
- The current quantum computer structure makes it difficult to ever build QC into devices such as mobile phones.

The processes of quantum physics are very sensitive to any kind of disturbance from the environment, such as heat, radiation and magnetic fields. For this reason, quantum computer chips are protected by several levels of shielding and cooled down almost to absolute zero (-273 °C). One or more quantum computers could be set up in today's computer centres and used on site or via the cloud.

Researchers expect that a development period ranging from five to ten years will be needed in order to build error-free quantum computers with a larger number of qubits and considerably longer computation periods.

In the next article in this series, we will delve into more details of quantum computing. END

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Disclaimer

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